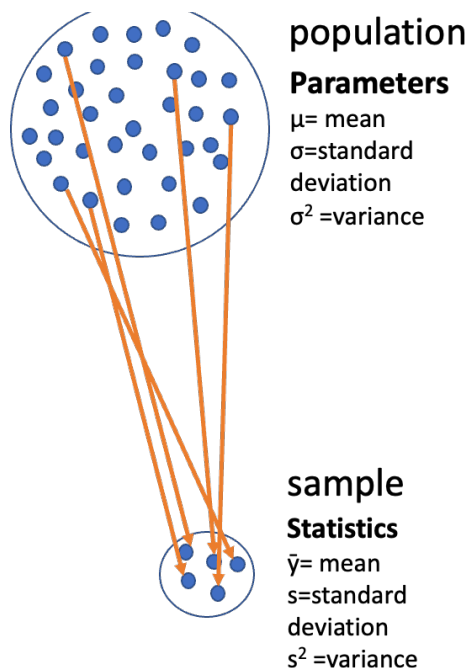


Lecture 05: Probability and Statistical Inference

Bill Perry

Lecture 4: Review

- Introduction to histograms or frequency distributions
- Probability Distribution Functions (PDF)
 - Z scores and T scores
- Tests of means using T-Tests
 - one sample
 - two sample



- Tests of means using T-Tests
 - one sample - is the sample mean different from a hypothesized mean?
 - two sample - are the sample means from two samples the same or different?
- for Two sample T-tests the $df = n_1 + n_2 - 2 = 8 + 8 - 2 = 14$

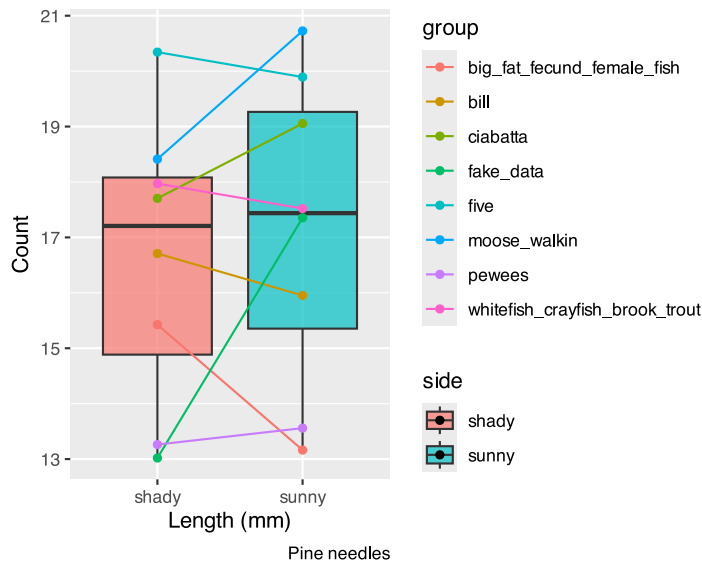
```
# A tibble: 2 × 5
  side mean_length sd_length se_length count
<chr>    <dbl>    <dbl>    <dbl> <int>
1 shady     17.6     2.51     0.886     8
2 sunny     16.2     2.64     0.934     8
```

Lecture 5: Probability and Statistical Inference

The goals for today

- Statistical inference fundamentals
- Hypothesis testing principles
- T Distributions

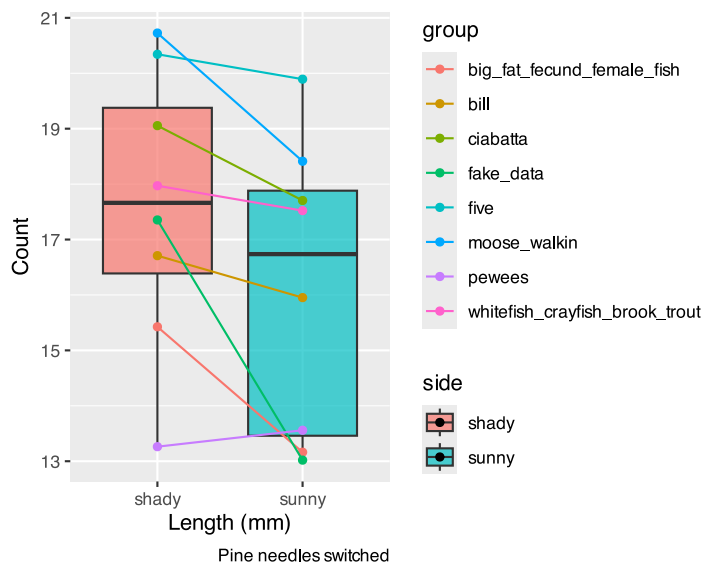
- One sample T Test
- Two sample T Test
- Paired T Test
- Assumption tests



Lecture 5: Probability and Statistical Inference

The goals for today

- Statistical inference fundamentals
- Hypothesis testing principles
- T Distributions
- One sample T Test
- Two sample T Test
- Paired T Test

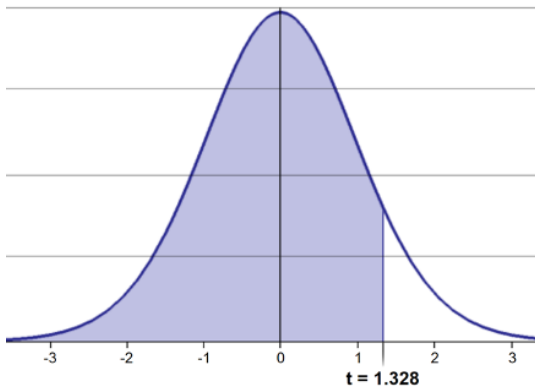


Lecture 5: One-tailed Questions

One-tailed questions: area of distribution left or (right) of a certain value for a one sample test

- $n=8$ ($df=7$) - 95% of the observations found left

- $t = 1.895$ (5% are outside)



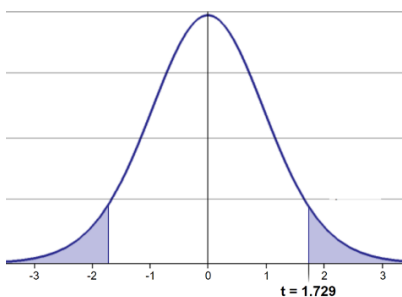
cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
Confidence Level											

XXXX

Lecture 5: Two-tailed Questions

Two-tailed questions refer to area between certain values

- $n = 8$ ($df = 7$), 95% of the observations are between
- $t = -2.365$ and $t = 2.365$ (2.5% are outside on each side)
- One tailed was $t = 1.895$ (5% are outside)



cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
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Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Lecture 5: Calculating CI Example

Let's calculate CIs again:

Use two-sided test

- $CI = \bar{y} \pm t \cdot \frac{s}{\sqrt{n}}$
- 95% CI Sample A: $= 17.6 \pm 2.365 \cdot (2.51/(8^{0.5})) = \pm 2.098746$
- The 95% CI is between 15.50 and 19.70
- "The 95% CI for the population mean from sample A is 17.6 ± 2.1

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
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	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Lecture 5: Applications of t-distribution

So:

- Can assess confidence that population mean is within a certain range
- Can use t distribution to ask questions like:
 - “What is probability of getting sample with mean = $\bar{y}$ from population with mean = μ ?” (1 sample t-test)
 - “What is the probability that two samples came from the same population?” (2 sample t-test)

Lecture 5: One Sample T-Test

We want to test if the mean needle length on one side differs from 15mm.

Activity: Define hypotheses and identify assumptions

$H_0: \mu = 15$ (The mean needle length on shade side is 15mm)

$H_1: \mu \neq 15$ (The mean needle length on shade side is not 240mm)

Assumptions for t-test:

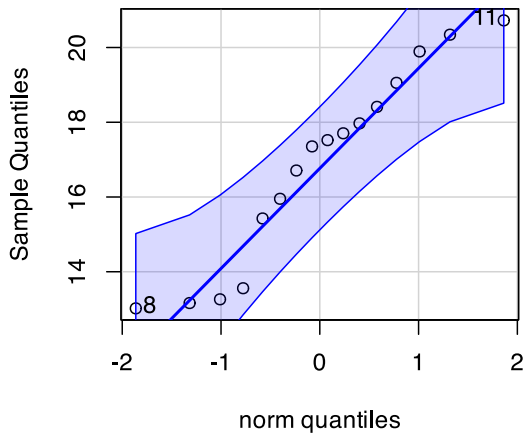
1. Data is normally distributed
2. Observations are independent
3. No significant outliers

Assumptions in R - qqplots from car

```
# YOUR TASK: Test normality of all pine needle lengths
# QQ Plot
qqPlot(ps_df$length_mm,
```

```
main = "QQ Plot for length of pine needles",
ylab = "Sample Quantiles")
```

QQ Plot for length of pine needles



```
[1] 8 11
```

Statistical Test of Normality

Shapiro-Wilk test

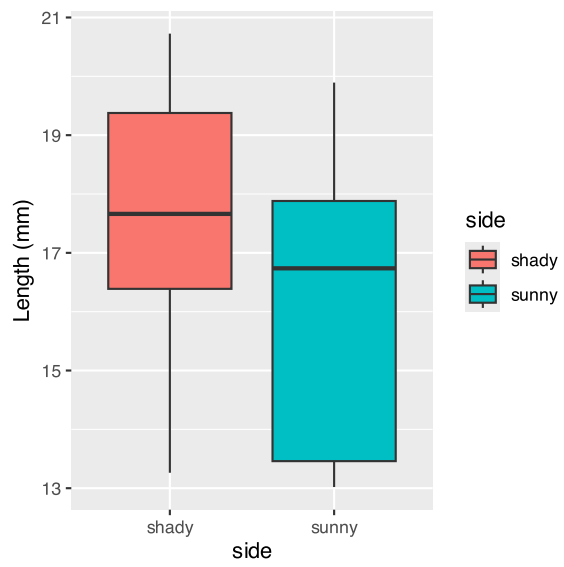
```
# Shapiro-Wilk test
shapiro.test(ps_df$length_mm)
```

Shapiro-Wilk normality test

data: ps_df\$length_mm
W = 0.92754, p-value = 0.2228

Checking for Outliers

```
# Check for outliers using boxplot
# YOUR CODE HERE
# Create a boxplot comparing the two lakes
shady_sunny_plot <- ps_df %>%
  ggplot(aes(x = side, y = length_mm, fill = side)) +
  geom_boxplot() +
  labs(
    x = "side",
    y = "Length (mm)",
    fill = "side")
shady_sunny_plot
```



Practice Exercise 1: One-Sample t-Test

💡 Practice Exercise 1: One-Sample t-Test

Let's perform a one-sample t-test to determine if the mean needle length on the shady side differs from 15 mm:

```
# what is the mean
ps_shade_mean <- mean(ps_shady_df$length_mm, na.rm = TRUE)
cat("Mean:", round(ps_shade_mean, 1), "mm\n")
```

Mean: 17.6 mm

```
# Perform a one-sample t-test
t_test_result <- t.test(ps_shady_df$length_mm, mu = 15)
t_test_result
```

One Sample t-test

```
data: ps_shady_df$length_mm
t = 2.9414, df = 7, p-value = 0.02167
alternative hypothesis: true mean is not equal to 15
95 percent confidence interval:
 15.51092 19.70030
sample estimates:
mean of x
 17.60561
```

Interpret this test result by answering these questions:

1. What was the null hypothesis?
2. What was the alternative hypothesis?
3. What does the p-value tell us?
4. Should we reject or fail to reject the null hypothesis at $\alpha = 0.05$?
5. What is the practical interpretation of this result for botanists?

Lecture 5: Hypothesis Testing Framework

Hypothesis testing is a systematic way to evaluate research questions using data.

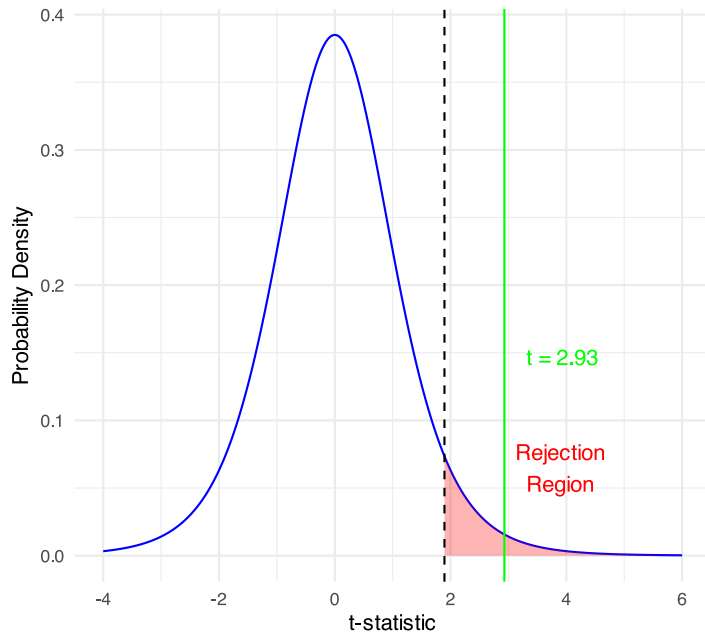
Key components:

1. **Null hypothesis (H_0):** Typically assumes “no effect” or “no difference”
2. **Alternative hypothesis (H_a):** The claim we're trying to support
3. **Statistical test:** Method for evaluating evidence against H_0
4. **P-value:** Probability of observing our results (or more extreme) if H_0 is true
5. **Significance level (α):** Threshold for rejecting H_0 , typically 0.05

Decision rule: Reject H_0 if p-value less than α or shorthand $p < 0.05$

One-Sample t-Test

Ho: $\mu = 15$ vs Ha: $\mu \neq 15$ ($\alpha = 0.05$, $df = 7$)



Lecture 5: Hypothesis Testing

Hypothesis testing is a systematic way to evaluate research questions using data.

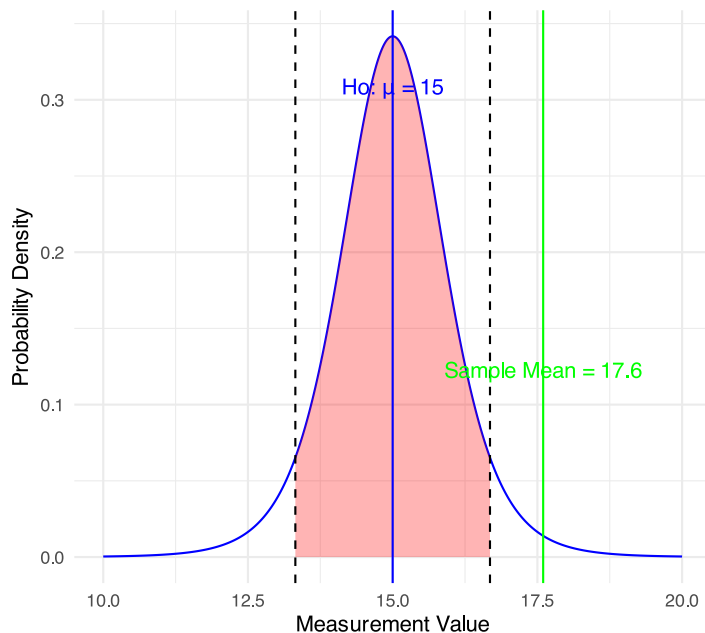
Key components:

1. **Null hypothesis (H_0):** Typically assumes “no effect” or “no difference”
2. **Alternative hypothesis (H_a):** The claim we’re trying to support
3. **Statistical test:** Method for evaluating evidence against H_0
4. **P-value:** Probability of observing our results (or more extreme) if H_0 is true
5. **Significance level (α):** Threshold for rejecting H_0 , typically 0.05

Decision rule: Reject H_0 if p-value $< \alpha$

One-Sample t-Test in Original Scale

Testing Ho: $\mu = 15$ ($\alpha = 0.05$, $df = 7$)



Lecture 5: Interpreting One-Sample T-Test Results

Activity: Interpret the t-test results

- What does the p-value tell us?
- Should we reject or fail to reject the null hypothesis?

How to report this result in a scientific paper:

“A one-sample t-test at $\alpha=0.05$ showed that the mean needle length (... mm, SD = ...) [was/was not] significantly different from the expected 15 mm, $t(\dots) = \dots$, $p = \dots$ ”

Lecture 5: Two Sample T-Tests Introduction

For example

- what is probability that population X is the same as population Y?
- How would you assess this question using what we learned?
- This is what we will do with the needle length again...



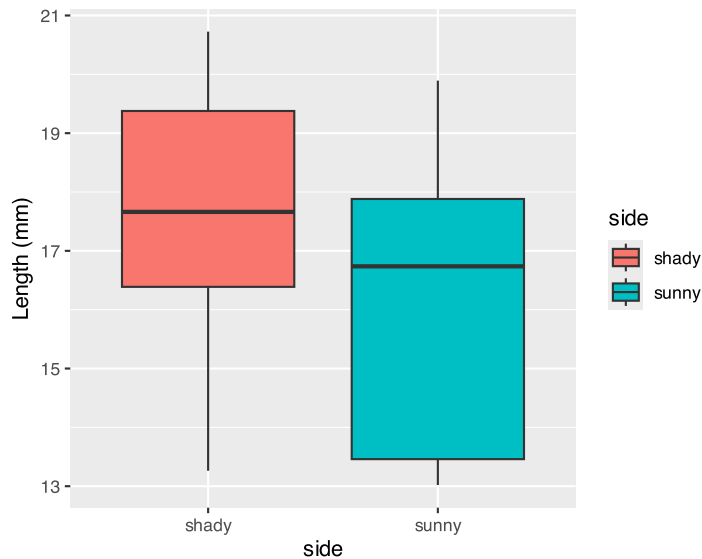
Lecture 5: Comparing Two Samples

For example

- what is probability that population X is the same as population Y?

How would you assess this question using what we learned?

```
shady_sunny_plot
```



```
# Based on the t-test results and the boxplot
#
# what can you conclude about the needle length on the two sides?
```

Practice Exercise 2: Formulating Hypotheses

💡 Practice Exercise 2: Formulating Hypotheses

For the following research questions about needle lengths write the null and alternative hypotheses:

1. Are needle lengths on shady and sunny sides different?

What are the hypotheses?

Ho =

Ha =

Lecture 5: Two-Sample T-Test Framework

Now, let's compare needles lengths from the two sides

Question: **Is there a significant difference in needle length between the sides?**

This requires a two-sample t-test.

Two-sample t-test compares means from two independent groups.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where:

- $\bar{x}_1$ and $\bar{x}_2$: These represent the sample means of the two groups you're comparing
- s_p^2 : This is the pooled variance, calculated as: $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$, where s_1^2 and s_2^2 are the sample variances of the two groups.
- **n_1 and n_2** : These are the sample sizes of the two groups.
- $\sqrt{(1/n_1 + 1/n_2)}$: This represents the pooled standard error.

$$t = \frac{SIGNAL}{NOISE}$$

Practice Exercise 3: Summary Statistics

💡 Practice Exercise 3: Calculate summary statistics grouped by lake

Before conducting the test, we need to understand the data for each group.

1. You need this and the graph to see what is going on

```
group_summary <- ps_df %>%
  group_by(side) %>%
  summarize(
    mean_length = mean(length_mm),
    sd_length = sd(length_mm),
    n = n(),
    se_length = sd_length / sqrt(n)
  )
group_summary
```

```
# A tibble: 2 × 5
  side mean_length sd_length    n se_length
<chr>    <dbl>    <dbl> <int>    <dbl>
1 shady      17.6      2.51     8     0.886
2 sunny      16.2      2.64     8     0.934
```

Practice Exercise 4: Effect Size

💡 Practice Exercise 4: Effect size

We could also look at the difference in means... some cool code here

```
# Assuming your dataframe is called df
group_summary %>%
  summarize(difference = mean_length[side == "shady"] - mean_length[side == "sunny"])
```

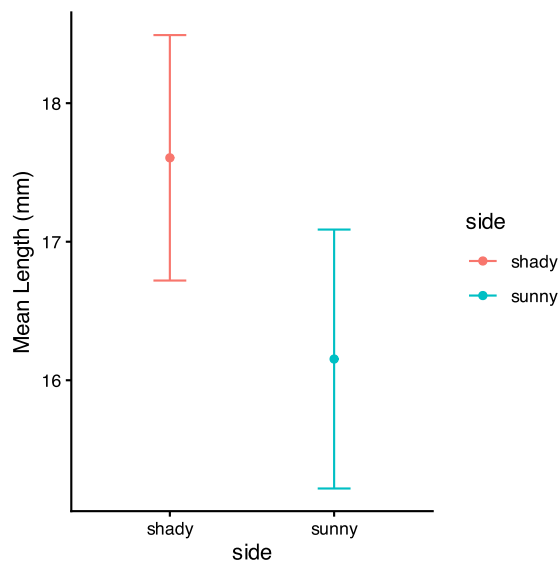
```
# A tibble: 1 × 1
  difference
    <dbl>
1      1.45
```

Practice Exercise 5: ggplot Summary Statistics

💡 Practice Exercise 5: Using GGPLOT to get summary plot

GGplot also has code to make the mean and standard error plots we are interested in along with a lot of others

```
# Assuming your dataframe is called df
needle_mean_se_plot <- ggplot(ps_df, aes(x = side, y = length_mm, color = side)) +
  stat_summary(fun = mean, geom = "point") +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) +
  labs(
    x = "side",
    y = "Mean Length (mm)" ) +
  theme_classic()
needle_mean_se_plot
```



Lecture 5: Testing Assumptions for Two-Sample T-Test

For a two-sample t-test, we need to check:

1. Normality within each group
2. Equal variances between groups (for standard t-test)
3. Independent observations

If assumptions are violated:

- Welch's t-test (unequal variances)
- Non-parametric alternatives (Mann-Whitney U test)

Practice Exercise 6: Separate Group Data

💡 Practice Exercise 7: Test normality of sunny pine needle lengths

Note you need to test each groups separately...

```
# how do you make separate dataframes to do this on?  
# Separate data by groups  
head(ps_shady_df)
```

```
# A tibble: 6 × 5  
# Groups:   group, tree_no, tree_char [6]  
  group          tree_no tree_char side length_mm  
  <chr>          <dbl> <chr>   <chr>    <dbl>  
1 big_fat_fecund_female_fish      2 tree_2  shady     15.4  
2 bill                        3 tree_3  shady     16.7  
3 ciabatta                     5 tree_5  shady     19.1  
4 fake_data                     8 tree_8  shady     17.4  
5 five                         1 tree_1  shady     20.3  
6 moose_walkin                   7 tree_7  shady     20.7
```

```
head(ps_sunny_df)
```

```
# A tibble: 6 × 5  
# Groups:   group, tree_no, tree_char [6]  
  group          tree_no tree_char side length_mm  
  <chr>          <dbl> <chr>   <chr>    <dbl>  
1 big_fat_fecund_female_fish      2 tree_2  sunny     13.2  
2 bill                        3 tree_3  sunny     16.0  
3 ciabatta                     5 tree_5  sunny     17.7  
4 fake_data                     8 tree_8  sunny     13.0  
5 five                         1 tree_1  sunny     19.9  
6 moose_walkin                   7 tree_7  sunny     18.4
```

Practice Exercise 8: Combined Normality Test

💡 Practice Exercise 8: Test Normality at one time

There are always a lot of ways to do this in R

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- ps_df %>%
  group_by(side) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
  )
normality_results
```

```
# A tibble: 2 × 4
  side shapiro_stat shapiro_p_value normal_distribution
<chr>   <dbl>         <dbl> <chr>
1 shady     0.966         0.868 Normal
2 sunny     0.900         0.289 Normal
```

Practice Exercise 13: Test Equal Variances

💡 Practice Exercise 13: Test equal variances

Levenes test can be done on the original dataframe

Note: the Levenes Test should be NOT SIGNIFICANT - What is the null hypothesis

```
# Method 1: Using car package's leveneTest
# This is often preferred as it's more robust to departures from normality
levene_result <- leveneTest(length_mm ~ side, data = ps_df)
print("Levene's Test for Homogeneity of Variance:")
```

```
[1] "Levene's Test for Homogeneity of Variance:"
```

```
print(levene_result)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

Lecture 5: Conducting the Two-Sample T-Test

Now we can compare the mean needle lengths between shady and sunny sides.

H₀: $\mu_1 = \mu_2$ (The needle lengths do not differ)

H_a: $\mu_1 \neq \mu_2$ (The mean needle lengths differ - direction is not specified)

Calculate t-statistic manually (optional) - YOUR CODE HERE:

$$t = (\text{mean1} - \text{mean2}) / \sqrt{(s1^2/n1) + (s2^2/n2)}$$

Deciding between:

- **Standard t-test (equal variances)**
- Welch's t-test (unequal variances)

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(t_test_result)
```

Two Sample t-test

```
data: length_mm by side
t = 1.1279, df = 14, p-value = 0.2783
alternative hypothesis: true difference in means between group shady and group sunny is not
equal to 0
95 percent confidence interval:
 -1.309330  4.214005
sample estimates:
mean in group shady mean in group sunny
      17.60561      16.15328
```

Lecture 5: Conducting the Two-Sample T-Test

Now we can compare the mean needle lengths between shady and sunny sides.

$H_0: \mu_1 = \mu_2$ (The needle lengths do not differ)

$H_a: \mu_1 \neq \mu_2$ (The mean needle lengths differ - direction is not specified)

Calculate t-statistic manually (optional) - YOUR CODE HERE:

$$t = (\text{mean1} - \text{mean2}) / \sqrt{(s1^2/n1) + (s2^2/n2)}$$

Deciding between:

- Standard t-test (equal variances)
- **Welches t-test (unequal variances)**

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = FALSE)
print("Welches two-sample t-test:")
```

```
[1] "Welches two-sample t-test:"
```

```
print(t_test_result)
```

```
Welch Two Sample t-test

data:  length_mm by side
t = 1.1279, df = 13.96, p-value = 0.2784
alternative hypothesis: true difference in means between group shady and group sunny is not
equal to 0
95 percent confidence interval:
 -1.310069  4.214743
sample estimates:
mean in group shady mean in group sunny
      17.60561      16.15328
```

Lecture 5: Difference between a Two-Sample T and Welch's T Test

Standard t-test (Student's t-test)

- **Assumes equal variances** between the two groups being compared
- Uses a **pooled variance estimate** that combines data from both group
- Has **higher statistical power** when the equal variance assumption is met
- **Degrees of freedom** = $n_1 + n_2 - 2$

Welch's t-test

- **Does not assume equal variances** between groups (also called the "unequal variances t-test")
- Uses **separate variance estimates** for each group
- More **robust** when group variances are different
- Uses a more complex **degrees of freedom calculation (Welch-Satterthwaite equation)** and decimal!!!
- **Degrees of freedom** are typically non-integer and usually smaller than the standard t-test

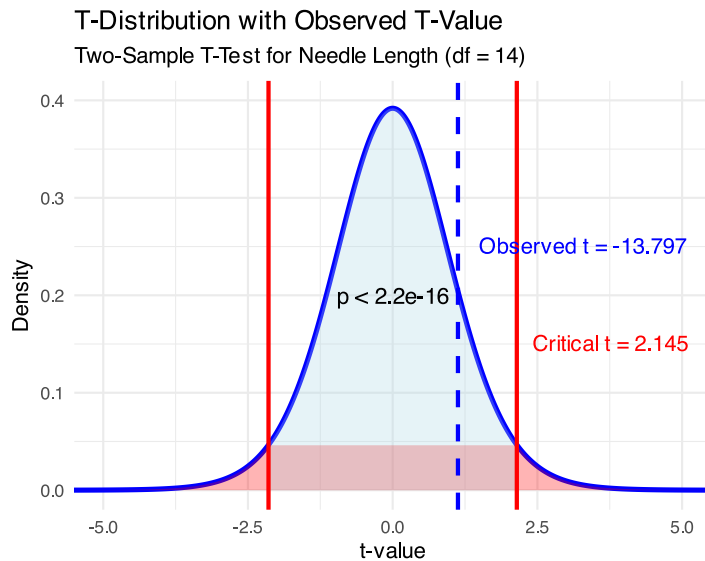
Lecture 5: Interpreting Two-Sample T-Test Results

Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on sunny vs shady sides?

How to report this result in a scientific paper:

"A two-tailed, two-sample t-test at $\alpha=0.05$ showed [a significant/no significant] difference in needle length between sunny ($M = \dots$, $SD = \dots$) and shady ($M = \dots$, $SD = \dots$) sides of pine trees, $t(\dots) = \dots$, $p = \dots$ "



Lecture 5: Now what does a paired T test tell us

Paired t-test:

Compares two measurements from the same subjects or matched pairs Tests whether the mean difference between paired observations equals zero Examples: before/after measurements on the same people, left vs right measurements, matched case-control studies Uses the differences between pairs as the data points Generally more powerful because it controls for individual variation

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

ps_wide_df <- ps_df %>%
  pivot_wider(
    names_from = "side",
    values_from = length_mm
  )

# Standard t-test (if variances are equal)
paired_t_test_result <- t.test(ps_wide_df$sunny, ps_wide_df$shady, paired = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(paired_t_test_result)
```

Paired t-test

```
data: ps_wide_df$sunny and ps_wide_df$shady
t = -2.7818, df = 7, p-value = 0.02723
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -2.6868652 -0.2178092
sample estimates:
```

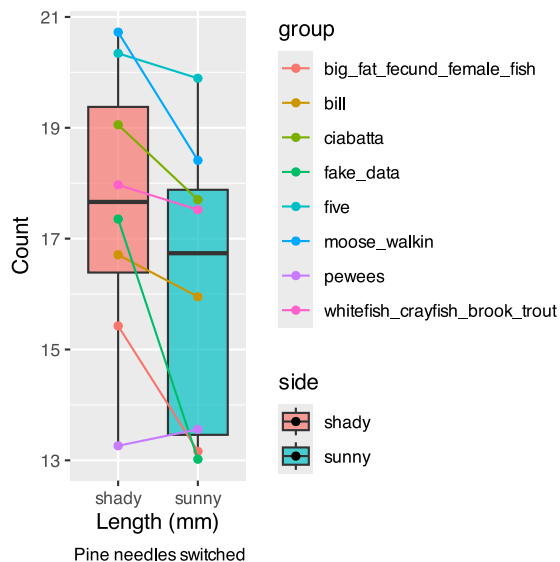
```
mean difference
-1.452337
```

```
# Welch's t-test (if variances are unequal)
# YOUR CODE HERE
```

Lecture 5: What is going on??

Note that there is a lot of variation within trees but the trend is the same

```
ps_plot
```



Lecture 5: Assumptions of Parametric Tests

Common assumptions for t-tests:

1. Normality: Data comes from normally distributed populations
2. Equal variances (for two-sample tests)
3. Independence: Observations are independent
4. No outliers: Extreme values can influence results

What can we do if our data violates these assumptions?

Alternatives when assumptions are violated:

- Data transformation (log, square root, etc.)
- Non-parametric tests
- Robust statistical methods

Lecture 5: Summary and Conclusions

In this activity, we've:

1. Formulated hypotheses about pine needle length
2. Tested assumptions for parametric tests
3. Conducted one-sample and two-sample t-tests
4. Visualized data using appropriate methods
5. Learned how to interpret and report t-test results

Key takeaways:

- Always check assumptions before conducting tests
- Visualize your data to understand patterns
- Report results comprehensively
- Consider alternatives when assumptions are violated - non parametric tests...